

KELT-23Ab

R-2109

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_260304 · created 2026-08-11T00:15:42+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461103.8332 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.01
Transit depth (Rp/Rs)^2	1.49 +/- 0.24 [%]
Orbital Inclination (inc)	85.74 +/- 0.79
Airmass coefficient 1 (a1)	1.2001 +/- 0.0062
Airmass coefficient 2 (a2)	-0.1369 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	0.52 %
Best Comparison Star	None
Optimal Aperture	4.46
Optimal Annulus	8.48
Transit Duration (day)	0.086 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.01 vs 0.132 (deviation 1.0σ)
Duration fit vs published	0.086 d vs 0.0954 d (deviation 2.48σ)
Mid-time O–C	-2.5 min
Depth SNR	12.2
Residual scatter	0.52 %

Light curve

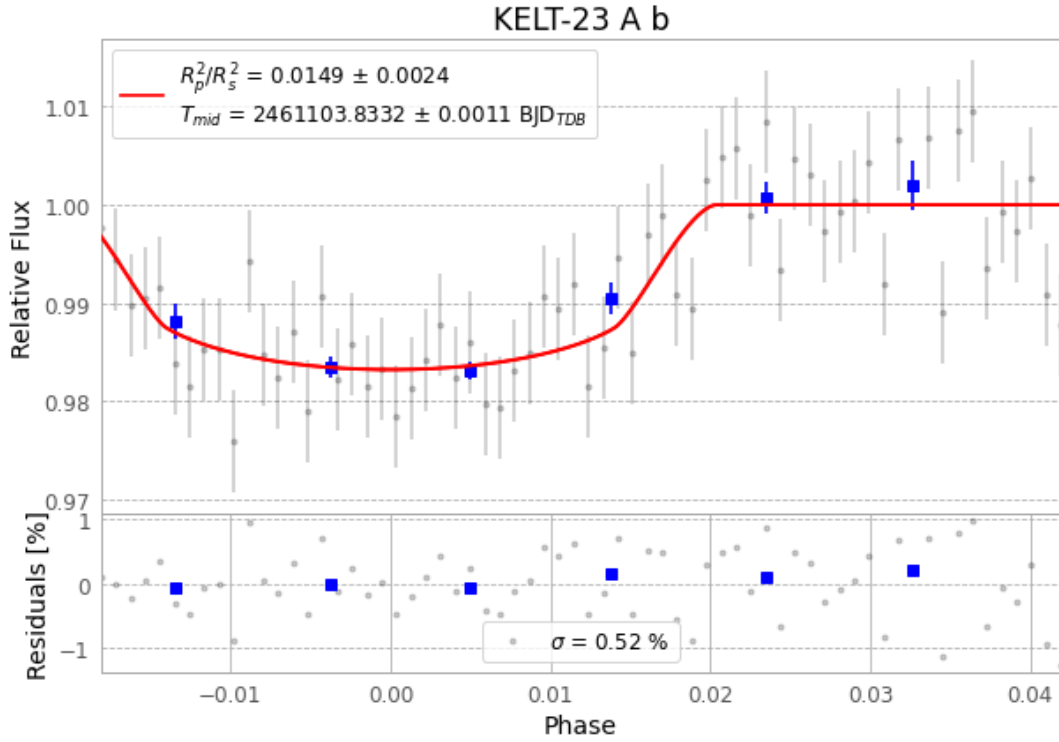


Figure 1 — FinalLightCurve_KELT-23 A b_2026-03-03.png (SHA-256 df1ecddc7a74cb84...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 143], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 78361e3fdce8e666...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), KELT-23A260304065715.FITS → KELT-23A260304101215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-260304.log (f50bacaa77b7...), FinalLightCurve_KELT-23 A b_2026-03-03.png (df1ecddc7a74...), FinalLightCurve_KELT-23 A b_2026-03-03.csv (b887e0416b7c...)

Compute resources

Wall time 177.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:15Z.